SELF-LEARNING MOBILE SECURITY SYSTEM USING FACE DETECTION AND BEHAVIOURAL ANALYSIS WITH AI

A PROJECT REPORT

Submitted by

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In partial fulfillment of the requirement for the award of the Degree of

BACHELOR OF SCIENCE

in

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UNIVERSITY OF MADRAS

Under the guidance of

Mrs.K.Abirami

ASSISTANT PROFESSOR



JEPPIAAR COLLEGE OF ARTS AND SCIENCE

Padur, Chennai-603103

2025– 2026

DECLARATION

We, **LOKESHWARAN G(222312110), SUDARSHAN S(222312117),GOKUL PRASATH S (222312107)** of III BSc. Computer Science with Artificial Intelligence, hereby declare that this project report on **"SELF-LEARNING MOBILE SECURITY SYSTEM USING FACE DETECTION AND BEHAVIOURAL ANALYSIS WITH AI"** submitted to University of Madras in partial fulfillment of the requirement for the award of the Degree of Bachelor of Computer science, under the guidance of **Mrs. K.Abirami** has not been submitted earlier to any other university or institute for the award of any degree.

Place: **(Lokeshwaran G)**

(Sudarshan S)

Date:

(Gokul Prasath S)

BONAFIDE CERTIFICATE

This is to certify that the project titled "**SELF-LEARNING MOBILE SECURITY SYSTEM USING FACE DETECTION AND BEHAVIOURAL ANALYSIS WITH AI**" is the bonafide work done by LOKESHWARAN G(222312110)**,** SUDARSHAN S(222312117)and GOKUL PRASATH S(222312107) final year students of Jeppiaar College of Arts and Science, Padur, Chennai in partial fulfillment of the requirement for the award of the Degree of Bachelor of Computer Science during the academic year 2025-2026.

PROJECT GUIDE

HEAD OF THE DEPARTMENT

Date:

VIVA VOICE EXAMINATION

This viva voice examination of the project titled "**SELF-LEARNING MOBILE SECURITY SYSTEM USING FACE DETECTION AND BEHAVIOURAL ANALYSIS WITH AI**" is submitted by LOKESHWARAN G(222312110)**.** SUDARSHAN S(222312117) and GOKUL PRASATH S(222312107) of III BSc. Computer Science with Artificial Intelligence. University of Madras Examination held on __.

INTERNALEXAMINER **EXTERNAL EXAMINER**

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Lokeshwaran G.

Sudarshan S.

Gokul prasth S.

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1. ABSTRACT:

The project “Self-Learning Mobile Security System Using Face Detection and Behavioural Analysis with AI” is designed to improve mobile security by using intelligent and adaptive authentication techniques. Traditional mobile security methods, such as passwords and PINs, can be easily forgotten, guessed, or hacked. To overcome this problem, this system uses face detection, location tracking, and sensor-based behaviour analysis to provide stronger protection.

Face detection is used as the primary authentication method. The system uses the mobile front camera to detect and recognize the authorized user’s face. If the face matches the stored data, the user is authenticated and allowed to access the application. This method is fast, secure, and contactless.

The system also monitors the user’s location. The system stores frequently visited locations such as home, college, or workplace. If the device is accessed from an unusual or unknown location, the system should lock the application.

The system also uses sensor-based behaviour analysis instead of full behavioural analysis. It collects data from mobile sensors such as the accelerometer and gyroscope to monitor motion patterns like how the user holds the phone, movement patterns, and device handling style. These motion patterns are slightly different for each user.

The self-learning feature allows the system to continuously update and adapt to the user’s face data, location patterns, and sensor motion behaviour. If any unusual activity is detected, such as an unknown face, unusual location, or abnormal motion pattern, the system will lock the application

2. INTRODUCTION:

In addition to face detection, the system also uses location tracking and sensor-based behavioral analysis to enhance security. The system monitors the user's frequently visited locations and checks whether the device is being accessed from a trusted or untrusted location. It also collects data from mobile sensors such as the accelerometer and gyroscope to analyze the user's device handling and motion patterns. These patterns are unique for each user and can help in identifying unauthorized access.

The self-learning feature of the system uses Artificial Intelligence (AI) to continuously learn and update the user's face data, location patterns, and sensor behavior over time. If any unusual activity is detected, such as an unknown face, unknown location, or abnormal motion pattern, the system can automatically lock the device or send an alert to the user. This makes the system more secure, intelligent, and reliable.

3. PROBLEM SPECIFICATION:

The rapid growth of smartphone usage, mobile devices have become a primary storage medium for sensitive personal and professional data such as banking details, personal photos, confidential documents, and communication records. Traditional mobile security methods like PINs, passwords, and pattern locks are widely used but have significant limitations. These methods are vulnerable to threats such as shoulder surfing, brute-force attacks, password guessing, and unauthorized access if the device is lost or stolen.

Although biometric authentication methods like fingerprint and face recognition have improved mobile security, they still rely on a one-time verification process. Once the device is unlocked, there is no continuous monitoring to ensure that the authorized user is still using the device. This creates a major security gap, as an unauthorized user can gain access after the initial authentication.

Furthermore, existing security systems do not adapt to user behavior or learn from usage patterns. They lack intelligence in identifying suspicious activities such as unusual touch patterns, abnormal app usage, or changes in interaction behavior. This makes it difficult to detect insider threats or unauthorized access in real time.

Another major issue is the lack of integration between facial recognition and behavioral analysis. Most systems focus on either biometric authentication or usage patterns, but not both together. As a result, they fail to provide a multi-layered security approach that can enhance accuracy and reliability.

3.1 PROBLEM STATEMENT:

The digital world, smartphones store highly sensitive personal and financial information, making them a prime target for unauthorized access. Traditional security mechanisms such as passwords, PINs, and pattern locks provide only a basic level of protection and are vulnerable to attacks like guessing, shoulder surfing, and theft. Even advanced biometric systems like fingerprint and face recognition offer only one-time authentication at the time of unlocking the device, without ensuring continuous user verification. Once a device is unlocked, there is no mechanism to detect whether the current user is the legitimate owner or an intruder. This lack of continuous authentication creates a major security loophole, allowing unauthorized users to access confidential data. Additionally, existing systems do not effectively utilize behavioral patterns such as typing speed, touch dynamics, and usage habits to identify anomalies.

3.2 PROBLEM MODULE

- **Weak Password/PIN Security**
  Passwords, PINs, and pattern locks can be guessed, hacked, or seen by others (shoulder surfing).
- **One-Time Authentication Problem**
  Even biometric systems like fingerprint/face unlock only verify once. After unlocking, there is **no continuous checking**, so anyone can use the phone.
- **No Continuous User Monitoring**
  Current systems cannot detect if the real owner is still using the device after unlocking.
- **No Behavioural Intelligence**
  Existing security does not analyse user behaviour like touch patterns, typing speed, phone handling, etc.
- **No Multi-Layer Security Integration**
  Most systems use only one method (face or password), but they don't combine **face detection + behavioural analysis + location tracking**.
- **Need for an AI Self-Learning System**

  Your module highlights the need for an **AI-based self-learning approach** that adapts to user face, location, and sensor behaviour over time.

SOFTWARE REQUIREMENTS AND LIBRARIES:

To ensure seamless development, training, and deployment of our Brain Tumor Detection System, we utilized a combination of powerful tools and essential libraries. Below is a detailed breakdown of the software environment and key dependencies involved in our project.

3.3 SOFTWARE REQUIREMENTS:

Our project follows a structured workflow that leverages the following key platforms and tools:

- Operating System: Windows 10/11, macOS, or Linux (Ubuntu recommended) for compatibility with machine learning tools. Programming Language: Python3.xformodeldevelopmentandimageprocessing.
- Development Environment: Jupyter Notebook, VSCode, or PyCharm for coding and debugging.

Libraries& Frameworks:

- OpenCV–Image processing and preprocessing.
- NumPy&Pandas–Data handling and processing.
- Scikit-learn–Machine learning mode implementation (KNN classifier).
- Database(Optional): SQL iteor MySQL for storing user signature records.
- Browser Compatibility: Supports Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari for web-based implementations

4. Hardware Requirements

- Processor

  Intel i3 or higher (Intel i5 recommended) for smooth execution of AI and image processing tasks.

- RAM

  Minimum 4 GB RAM (8 GB recommended for better performance).

- Storage

  At least 20 GB free disk space for storing datasets, libraries, and application files.

- Camera

  Front camera (mobile or webcam) required for face detection and authentication.

- Sensors

  Mobile sensors such as accelerometer and a gyroscope for behaviour analysis.

- GPS Module

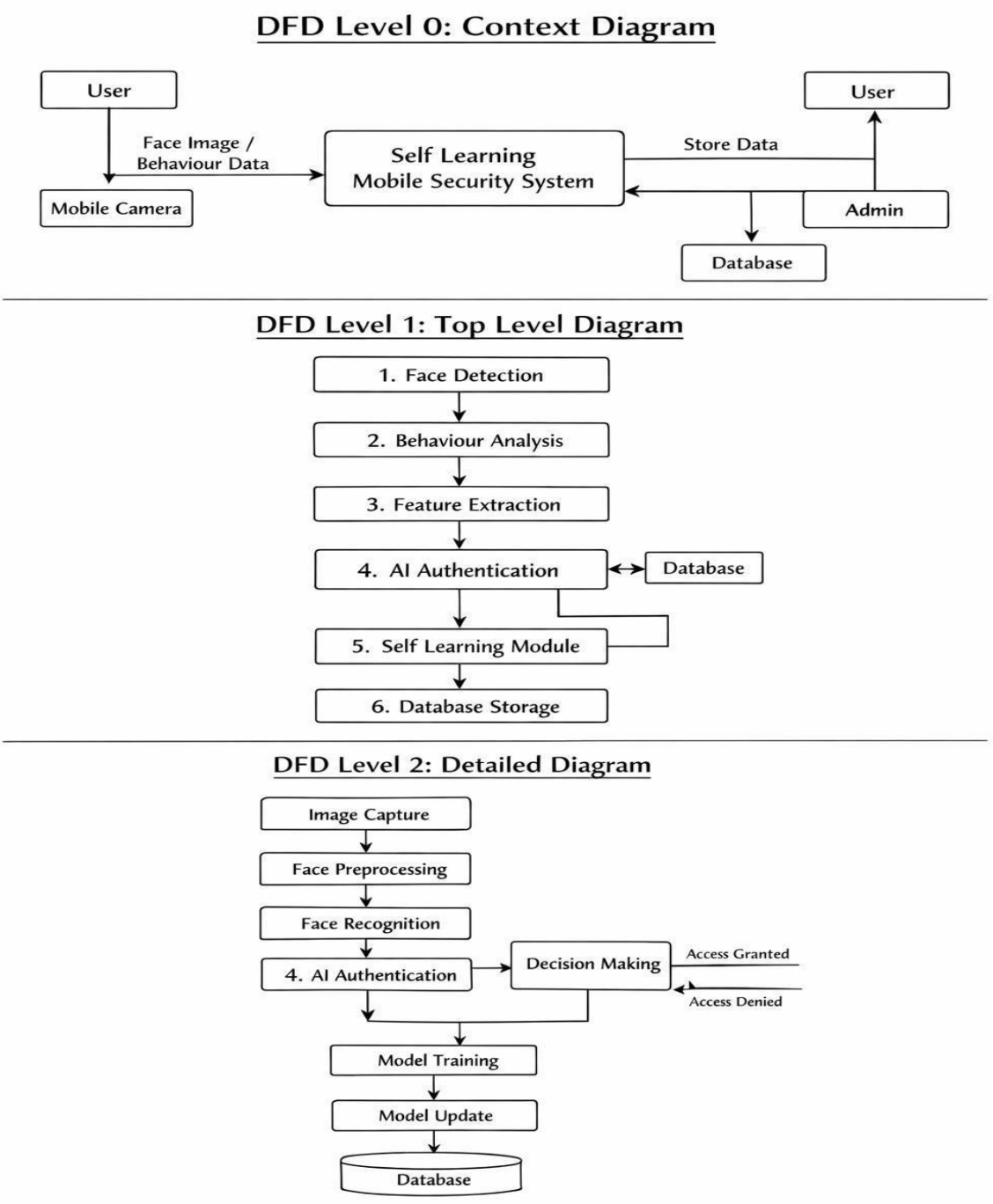
  Required for tracking user location and verifying trusted locations.

- Internet Connection

  Needed for installing libraries, updates, and optional cloud-based processing.

5. DATA FLOW DIAGRAM:

Self-Learning Mobile Security System Using Face Detection and Behavioural Analysis with Al



6. CODE:

FRONTEND:

```
<!DOCTYPE html>
<html>
<head>
  <!--
    If you are serving your web app in a path other than the root, change the
    href value below to reflect the base path you are serving from.

    The path provided below has to start and end with a slash "/" in order for
    it to work correctly.

    For more details:
    * https://developer.mozilla.org/en-US/docs/Web/HTML/Element/base

    This is a placeholder for base href that will be replaced by the value of
    the --base-href argument provided to flutter build.
  -->
  <base href="$FLUTTER_BASE_HREF">

  <meta charset="UTF-8">
  <meta content="IE=Edge" http-equiv="X-UA-Compatible">
  <meta name="description" content="A new Flutter project.">

  <!-- iOS meta tags & icons -->
  <meta name="mobile-web-app-capable" content="yes">
  <meta name="apple-mobile-web-app-status-bar-style" content="black">
  <meta name="apple-mobile-web-app-title" content="self_learning_mobile_security">
  <link rel="apple-touch-icon" href="icons/Icon-192.png">

  <!-- Favicon -->
  <link rel="icon" type="image/png" href="favicon.png"/>

  <title>self_learning_mobile_security</title>
```

```
  <link rel="manifest" href="manifest.json">
 </head>
 <body>
  <!--
    You can customize the "flutter_bootstrap.js" script.
    This is useful to provide a custom configuration to the Flutter loader
    or to give the user feedback during the initialization process.

    For more details:
    * https://docs.flutter.dev/platform-integration/web/initialization
   -->
  <script src="flutter_bootstrap.js" async></script>
 </body>
</html>
```

BACKEND:

```
{
    "name": "self_learning_mobile_security",
    "short_name": "self_learning_mobile_security",
    "start_url": ".",
    "display": "standalone",
    "background_color": "#0175C2",
    "theme_color": "#0175C2",
    "description": "A new Flutter project.",
    "orientation": "portrait-primary",
    "prefer_related_applications": false,
    "icons": [
        {
            "src": "icons/Icon-192.png",
            "sizes": "192x192",
            "type": "image/png"
        },
        {
            "src": "icons/Icon-512.png",
            "sizes": "512x512",
```

```
            "type": "image/png"
        },
        {
            "src": "icons/Icon-maskable-192.png",
            "sizes": "192x192",
            "type": "image/png",
            "purpose": "maskable"
        },
        {
            "src": "icons/Icon-maskable-512.png",
            "sizes": "512x512",
            "type": "image/png",
            "purpose": "maskable"
        }
    ]
}
```

Web_plugin_registant.dart

```
// Flutter web plugin registrant file.

// Generated file. Do not edit.

// @dart = 2.13
// ignore_for_file: type=lint

import 'package:camera_web/camera_web.dart';
import 'package:geolocator_web/geolocator_web.dart';
import 'package:package_info_plus/src/package_info_plus_web.dart';
import 'package:permission_handler_html/permission_handler_html.dart';
import 'package:sensors_plus/src/sensors_plus_web.dart';
import 'package:shared_preferences_web/shared_preferences_web.dart';
import 'package:wakelock_plus/src/wakelock_plus_web_plugin.dart';
import 'package:flutter_web_plugins/flutter_web_plugins.dart';

void registerPlugins([final Registrar? pluginRegistrar]) {
  final Registrar registrar = pluginRegistrar ?? webPluginRegistrar;
  CameraPlugin.registerWith(registrar);
```

```
      GeolocatorPlugin.registerWith(registrar);
      PackageInfoPlusWebPlugin.registerWith(registrar);
      WebPermissionHandler.registerWith(registrar);
      WebSensorsPlugin.registerWith(registrar);
      SharedPreferencesPlugin.registerWith(registrar);
      WakelockPlusWebPlugin.registerWith(registrar);
      registrar.registerMessageHandler()
```

package_config

```
{
  "configVersion": 2,
  "packages": [
    {
      "name": "_fe_analyzer_shared",
      "rootUri":
"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/_fe_analyzer_shared-67.0.0",
      "packageUri": "lib/",
      "languageVersion": "3.0"
    },
    {
      "name": "analyzer",
      "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/analyzer-6.4.1",
      "packageUri": "lib/",
      "languageVersion": "3.0"
    },
    {
      "name": "archive",
      "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/archive-4.0.9",
      "packageUri": "lib/",
      "languageVersion": "3.0"
    },
    {
      "name": "args",
      "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/args-2.7.0",
      "packageUri": "lib/",
```

```
      "languageVersion": "3.3"
    },
    {
      "name": "async",
      "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/async-2.13.0",
      "packageUri": "lib/",

"packageUri": "lib/",
"languageVersion": "3.0"
    },
    {
      "name": "build_runner",
      "rootUri":     "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/build_runner-
 2.4.13",
      "packageUri": "lib/",
      "languageVersion": "3.5"
    },
    {
      "name": "build_runner_core",
      "rootUri":
 "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/build_runner_core-7.3.2",
      "packageUri": "lib/",
      "languageVersion": "3.5"
    },
```

```
  {
   "name": "built_collection",
   "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/built_collection-
5.1.1",
   "packageUri": "lib/",
   "languageVersion": "2.12"
  },
  {
   "name": "built_value",
   "rootUri":       "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/built_value-
8.12.4",
   "packageUri": "lib/",
   "languageVersion": "3.0"
  },
  {
   "name": "camera",
   "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/camera-0.11.4",
   "packageUri": "lib/",
   "languageVersion": "3.9"
  },
  {
   "name": "camera_android_camerax",
   "rootUri":
"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/camera_android_camerax-
0.6.30",
   "packageUri": "lib/",
   "languageVersion": "3.9"
  },
```

"rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/camera-0.11.4",
"packageUri": "lib/",
"languageVersion": "3.9"
},
{
"name": "camera_android_camerax",
"rootUri":
"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/camera_android_camerax-0.6.30",
"packageUri": "lib/",
"languageVersion": "3.9"
},
{
"name": "camera_avfoundation",
"rootUri":
"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/camera_avfoundation-0.9.23+2",
"packageUri": "lib/",
"languageVersion": "3.9"
},
{
"name": "camera_platform_interface",
"rootUri":
"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/camera_platform_interface-2.12.0",
"packageUri": "lib/",
"languageVersion": "3.7"
},
{
"name": "camera_web",
"rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/camera_web-0.3.5+3",
"packageUri": "lib/",

```
    "languageVersion": "3.8"
   },
   {
    "name": "characters",
    "rootUri":          "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/characters-
1.4.1",
    "packageUri": "lib/",
    "languageVersion": "3.4"
   },
   {
    "name": "checked_yaml",
    "rootUri":  "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/checked_yaml-
2.0.4",
    "packageUri": "lib/",
    "languageVersion": "3.8"
   },
   {
    "name": "clock",
    "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/clock-1.1.2",
    "packageUri": "lib/",
    "languageVersion": "3.4"
   },
   {
    "name": "code_assets",
    "rootUri":       "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/code_assets-
1.0.0",
    "packageUri": "lib/",
    "languageVersion": "3.9"
   },
   {
    "name": "code_builder",
    "rootUri":     "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/code_builder-
4.11.1",
```

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    "packageUri": "lib/",
    "languageVersion": "3.7"
   },
   {
    "name": "collection",
    "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/collection-1.19.1",
    "packageUri": "lib/",
    "languageVersion": "3.4"
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    "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/convert-3.1.2",
    "packageUri": "lib/",
    "languageVersion": "3.4"
   },
   {
    "name": "cross_file",
    "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/cross_file-0.3.5+2",
    "packageUri": "lib/",
    "languageVersion": "3.8"
   },
   {
    "name": "crypto",
    "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/crypto-3.0.7",
    "packageUri": "lib/",
    "languageVersion": "3.4"
   },
   {
    "name": "dart_style",
    "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/dart_style-2.3.6",
```

```
    "packageUri": "lib/",
    "languageVersion": "3.0"
   },
   {
    "name": "dbus",
    "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/dbus-0.7.12",
    "packageUri": "lib/",
    "languageVersion": "2.17"
   },
   {
    "name": "fake_async",
    "rootUri":        "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/fake_async-
1.3.3",
    "packageUri": "lib/",
    "languageVersion": "3.3"
   },
   {
    "name": "ffi",
    "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/ffi-2.2.0",
    "packageUri": "lib/",
    "languageVersion": "3.7"
   },
   {
    "name": "file",
    "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/file-7.0.1",
    "packageUri": "lib/",
    "languageVersion": "3.0"
   },
   {
    "name": "fixnum",
    "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/fixnum-1.1.1",
    "packageUri": "lib/",
    "languageVersion": "3.1"
```

```
    },
    {
      "name": "flutter",
      "rootUri": "file:///C:/flutter/packages/flutter",
      "packageUri": "lib/",
      "languageVersion": "3.9"
    },
    {
      "name": "flutter_foreground_task",
      "rootUri":
"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/flutter_foreground_task-
6.5.0",
      "packageUri": "lib/",
      "languageVersion": "2.18"
    },
    {
      "name": "flutter_plugin_android_lifecycle",
      "rootUri":
"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/flutter_plugin_android_lifecyc
le-2.0.33",
      "packageUri": "lib/",
      "languageVersion": "3.9"
    },
    {
      "name": "flutter_test",
      "rootUri": "file:///C:/flutter/packages/flutter_test",
      "packageUri": "lib/",
      "languageVersion": "3.9"
    },
    {
      "name": "flutter_web_plugins",
      "rootUri": "file:///C:/flutter/packages/flutter_web_plugins",
      "packageUri": "lib/",
```

```
    "languageVersion": "3.9"
   },
   {
    "name": "frontend_server_client",
    "rootUri":
"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/frontend_server_client-4.0.0",
    "packageUri": "lib/",
    "languageVersion": "3.0"
   },
   {
    "name": "geolocator",
    "rootUri":  "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/geolocator-
12.0.0",
    "packageUri": "lib/",
    "languageVersion": "2.15"
   },
   {
    "name": "geolocator_android",
    "rootUri":
"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/geolocator_android-4.6.2",
    "packageUri": "lib/",
    "languageVersion": "3.5"
   },
   {
    "name": "geolocator_apple",
    "rootUri":
"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/geolocator_apple-2.3.13",
    "packageUri": "lib/",
    "languageVersion": "3.5"
   },
   {
    "name": "geolocator_platform_interface",
    "rootUri":
```

"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/geolocator_platform_interface-4.2.6",
"packageUri": "lib/",
"languageVersion": "3.5"
},
{
"name": "geolocator_web",
"rootUri":
"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/geolocator_web-4.1.3",
"packageUri": "lib/",
"languageVersion": "3.5"
},
{
"name": "geolocator_windows",
"rootUri":
"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/geolocator_windows-0.2.5",
"packageUri": "lib/",
"languageVersion": "3.5"
},
{
"name": "glob",
"rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/glob-2.1.3",
"packageUri": "lib/",
"languageVersion": "3.3"
},
{
"name": "google_mlkit_commons",
"rootUri":
"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/google_mlkit_commons-0.9.0",
"packageUri": "lib/",
"languageVersion": "2.16"
},

```
    {
      "name": "google_mlkit_face_detection",
      "rootUri":
"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/google_mlkit_face_detection-
0.12.0",
      "packageUri": "lib/",
      "languageVersion": "2.16"
    },
    {
      "name": "graphs",
      "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/graphs-2.3.2",
      "packageUri": "lib/",
      "languageVersion": "3.4"
    },
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      "name": "hive",
      "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/hive-2.2.3",
      "packageUri": "lib/",
      "languageVersion": "2.12"
    },
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      "name": "hive_flutter",
      "rootUri":       "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/hive_flutter-
1.1.0",
      "packageUri": "lib/",
      "languageVersion": "2.12"
    },
    {
      "name": "hive_generator",
      "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/hive_generator-
2.0.1",
      "packageUri": "lib/",
      "languageVersion": "2.12"
```

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    },
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      "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/hooks-1.0.2",
      "packageUri": "lib/",
      "languageVersion": "3.10"
    },
    {
      "name": "http",
      "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/http-1.6.0",
      "packageUri": "lib/",
      "languageVersion": "3.4"
    },
    {
      "name": "http_multi_server",
      "rootUri":
"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/http_multi_server-3.2.2",
      "packageUri": "lib/",
      "languageVersion": "3.2"
    },
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      "name": "http_parser",
      "rootUri":        "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/http_parser-
4.1.2",
      "packageUri": "lib/",
      "languageVersion": "3.4"
    },
    {
      "name": "image",
      "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/image-4.8.0",
      "packageUri": "lib/",
      "languageVersion": "3.0"
    },
```

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      "packageUri": "lib/",
      "languageVersion": "3.4"
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      "name": "js",
      "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/js-0.7.2",
      "packageUri": "lib/",
      "languageVersion": "3.7"
    },
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      "rootUri":
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      "packageUri": "lib/",
      "languageVersion": "3.9"
    },
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      "name": "leak_tracker",
      "rootUri":      "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/leak_tracker-
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      "packageUri": "lib/",
      "languageVersion": "3.2"
    },
    {
      "name": "leak_tracker_flutter_testing",
      "rootUri":
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3.0.10",
      "packageUri": "lib/",
      "languageVersion": "3.2"
```

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      "packageUri": "lib/",
      "languageVersion": "3.2"
    },
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      "name": "logging",
      "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/logging-1.3.0",
      "packageUri": "lib/",
      "languageVersion": "3.4"
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      "rootUri":
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0.13.0",
      "packageUri": "lib/",
      "languageVersion": "3.5"
    },
    {
      "name": "meta",
      "rootUri": "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/meta-1.17.0",
      "packageUri": "lib/",
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    "languageVersion": "3.5"
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    "packageUri": "lib/",
    "languageVersion": "3.2"
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    "rootUri":
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    "packageUri": "lib/",
    "languageVersion": "3.10"
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    "name": "objective_c",
    "rootUri":      "file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/objective_c-
9.3.0",
    "packageUri": "lib/",
    "languageVersion": "3.10"
   },
   {
    "name": "package_config",
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2.2.0",
    "packageUri": "lib/",
    "languageVersion": "3.4"
   },
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    "name": "package_info_plus",
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    "languageVersion": "3.3"
   },
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    "name": "package_info_plus_platform_interface",
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nterface-3.2.1",
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    "languageVersion": "2.18"
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    "packageUri": "lib/",
    "languageVersion": "3.4"
   },
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2.1.5",
    "packageUri": "lib/",
    "languageVersion": "3.4"
   },
   {
    "name": "path_provider_android",
    "rootUri":
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2.2.22",
    "packageUri": "lib/",
    "languageVersion": "3.9"
   },
   {
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2.6.0",
      "packageUri": "lib/",
      "languageVersion": "3.10"
    },
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      "name": "path_provider_linux",
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      "packageUri": "lib/",
      "languageVersion": "2.19"
    },
    {
      "name": "path_provider_platform_interface",
      "rootUri":
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ace-2.1.2",
      "packageUri": "lib/",
      "languageVersion": "3.0"
    },
    {
      "name": "path_provider_windows",
      "rootUri":
"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/path_provider_windows-
2.3.0",
      "packageUri": "lib/",
      "languageVersion": "3.2"
    },
    {
      "name": "permission_handler",
      "rootUri":
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"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/permission_handler-11.4.0",
"packageUri": "lib/",
"languageVersion": "3.5"
},
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"name": "permission_handler_android",
"rootUri":
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"languageVersion": "3.5"
},
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"name": "permission_handler_apple",
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"file:///C:/Users/acer/AppData/Local/Pub/Cache/hosted/pub.dev/permission_handler_apple-9.4.7",
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"packageUri": "lib/",
"languageVersion": "3.3"
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    "languageVersion": "3.0"
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    "languageVersion": "3.6"
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    "languageVersion": "2.17"
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    "name": "sensors_plus",
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"languageVersion": "3.9"
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"name": "shared_preferences_foundation",
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    "languageVersion": "3.3"
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    "packageUri": "lib/",
    "languageVersion": "3.2"
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    "name": "shared_preferences_web",
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    "languageVersion": "3.4"
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    "name": "shared_preferences_windows",
    "rootUri":
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2.4.1",
    "packageUri": "lib/",
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      "languageVersion": "3.4"
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      "packageUri": "lib/",
      "languageVersion": "3.3"
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      "languageVersion": "3.9"
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      "languageVersion": "3.0"
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      "packageUri": "lib/",
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    "languageVersion": "3.4"
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    "packageUri": "lib/",
    "languageVersion": "3.1"
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    "languageVersion": "3.4"
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"languageVersion": "3.1"
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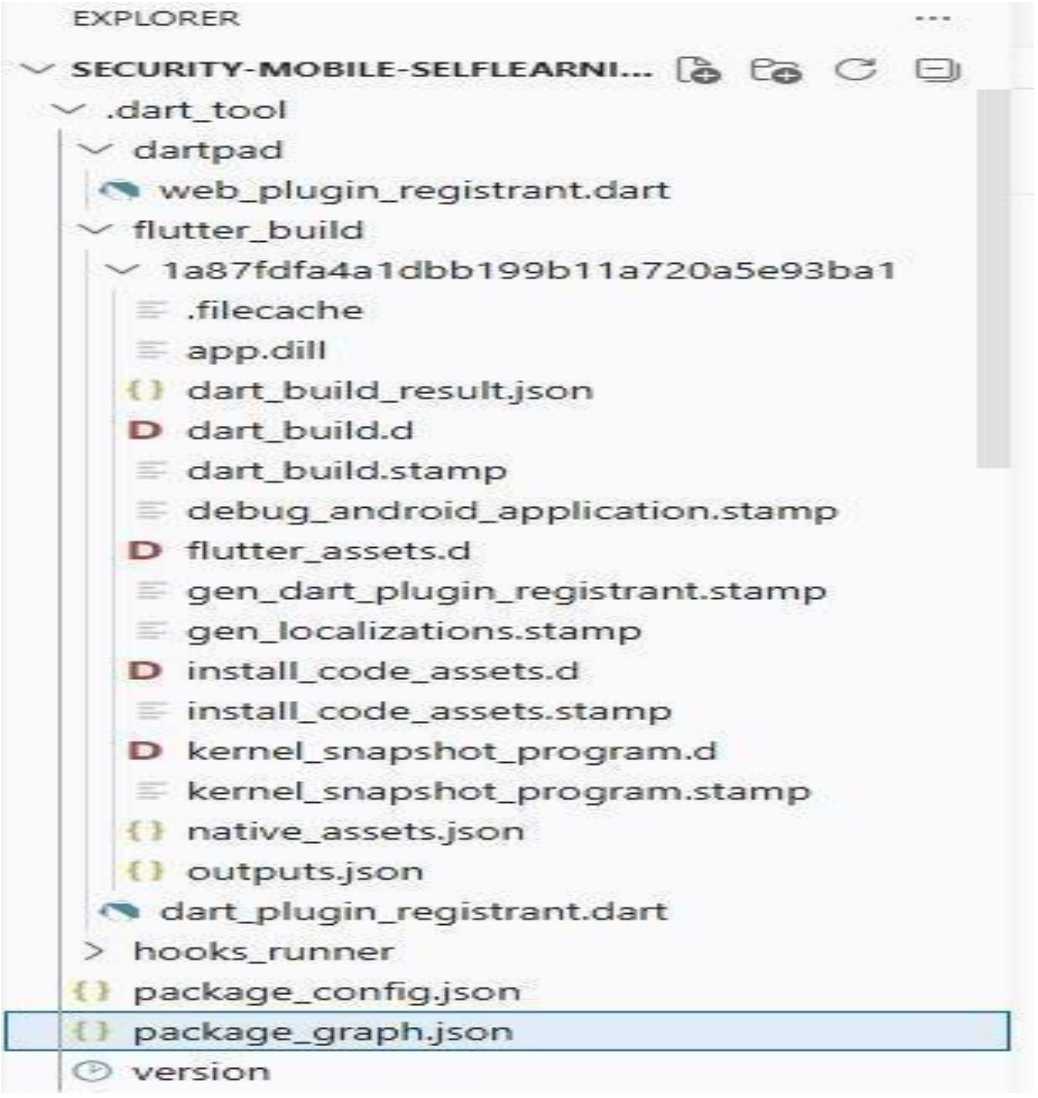
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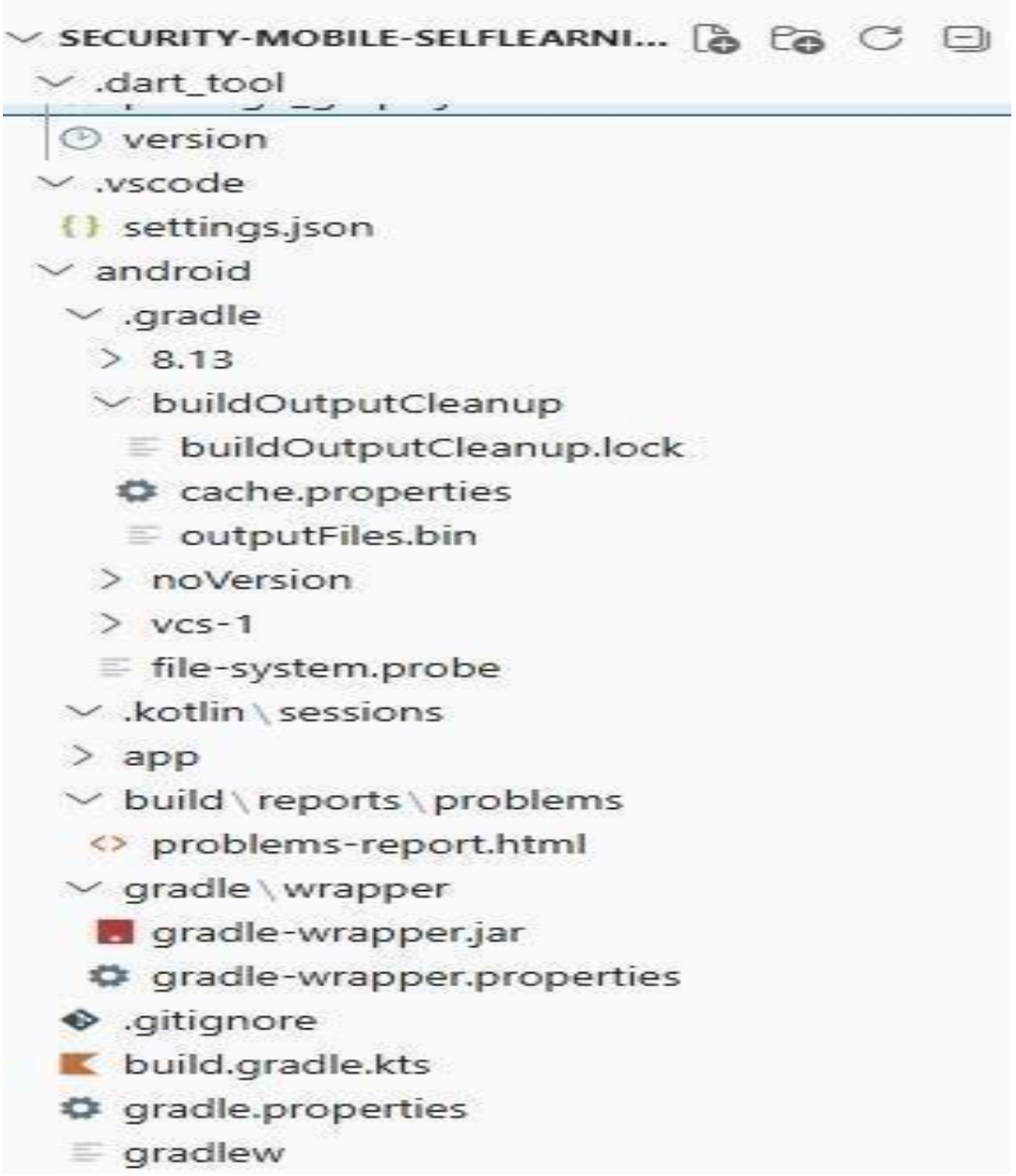
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    "languageVersion": "3.4"
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    "languageVersion": "3.3"
   },
   {
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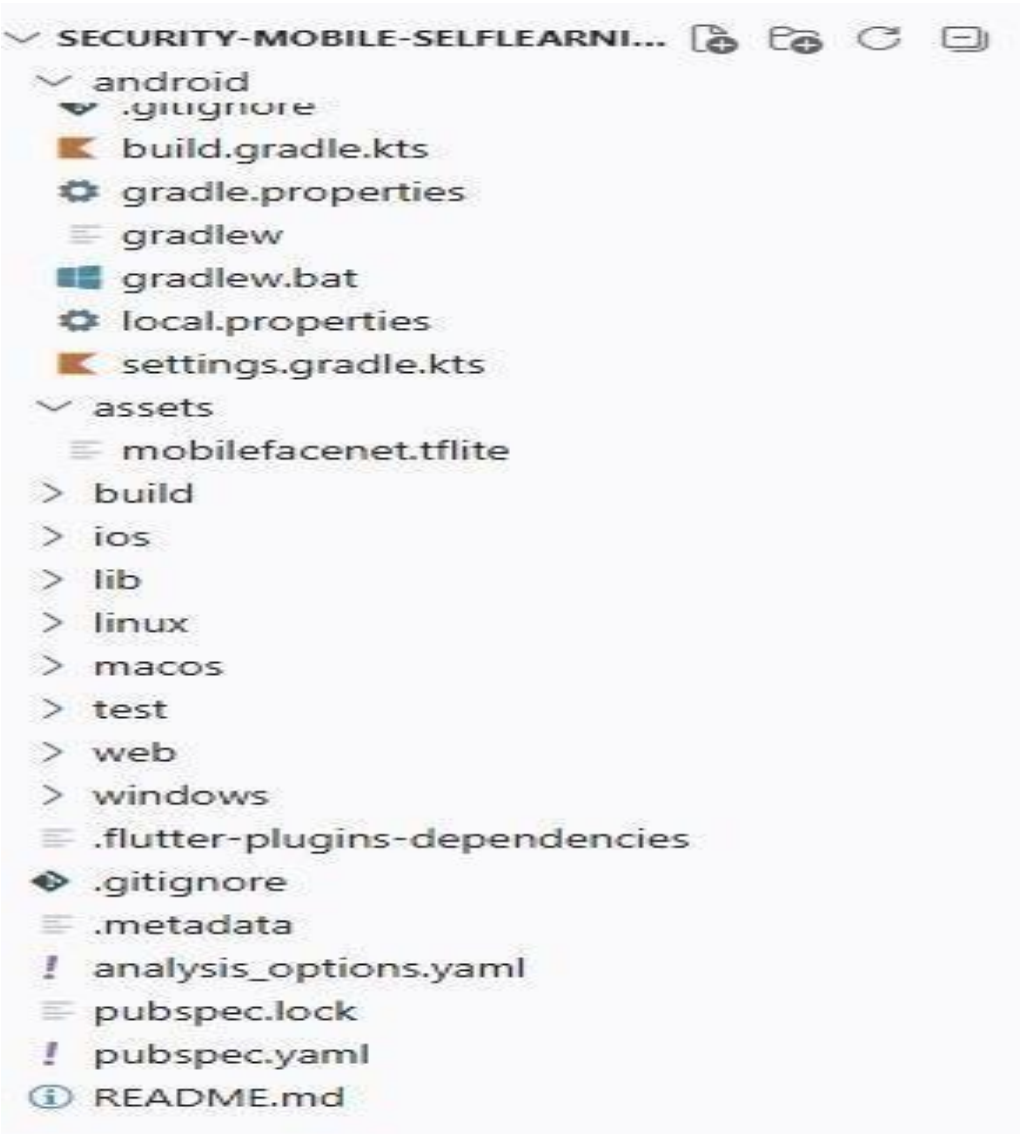
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    "languageVersion": "3.3"
   },
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    "languageVersion": "3.8"
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    "languageVersion": "3.4"
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```

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FOLDER

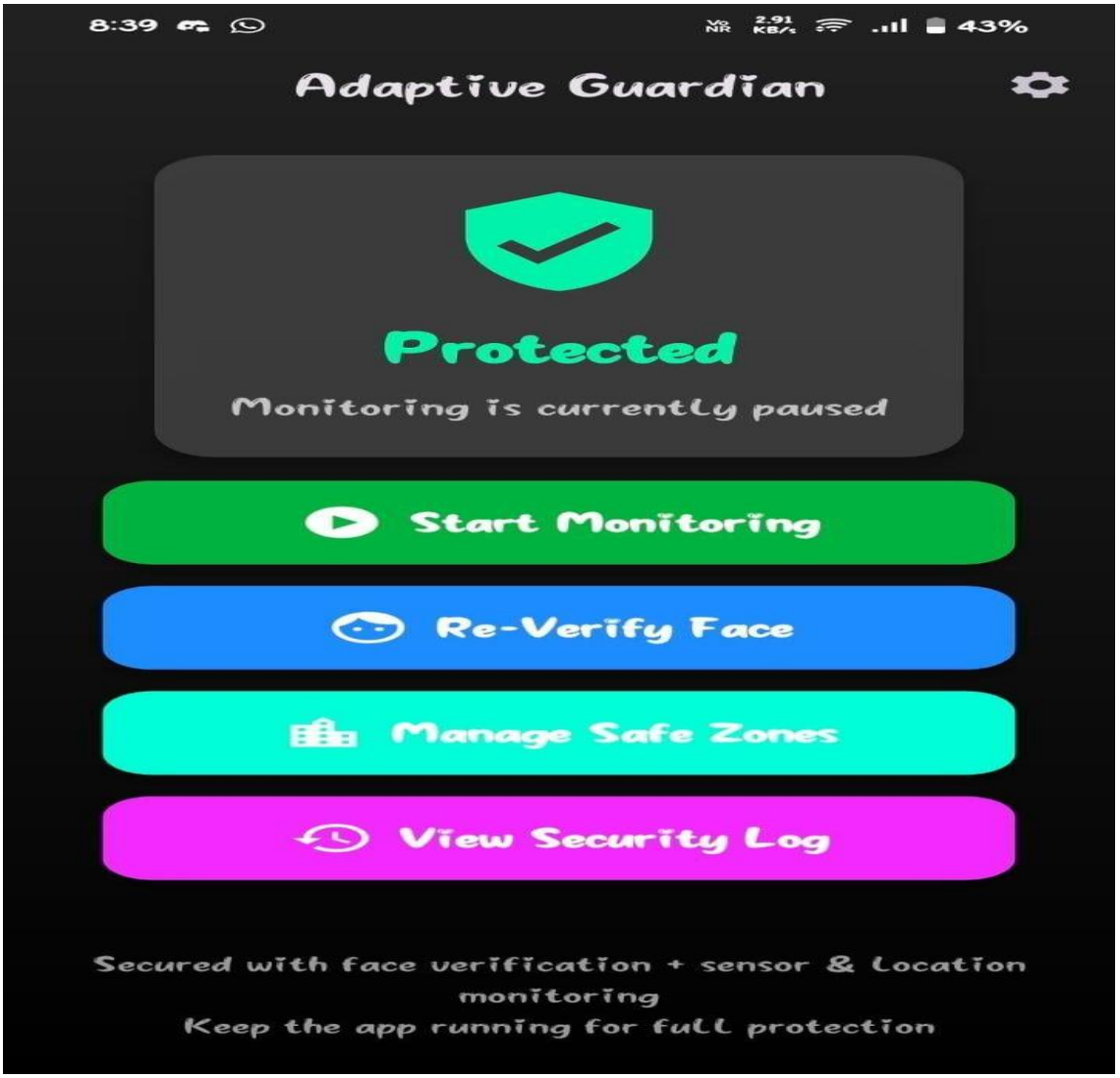


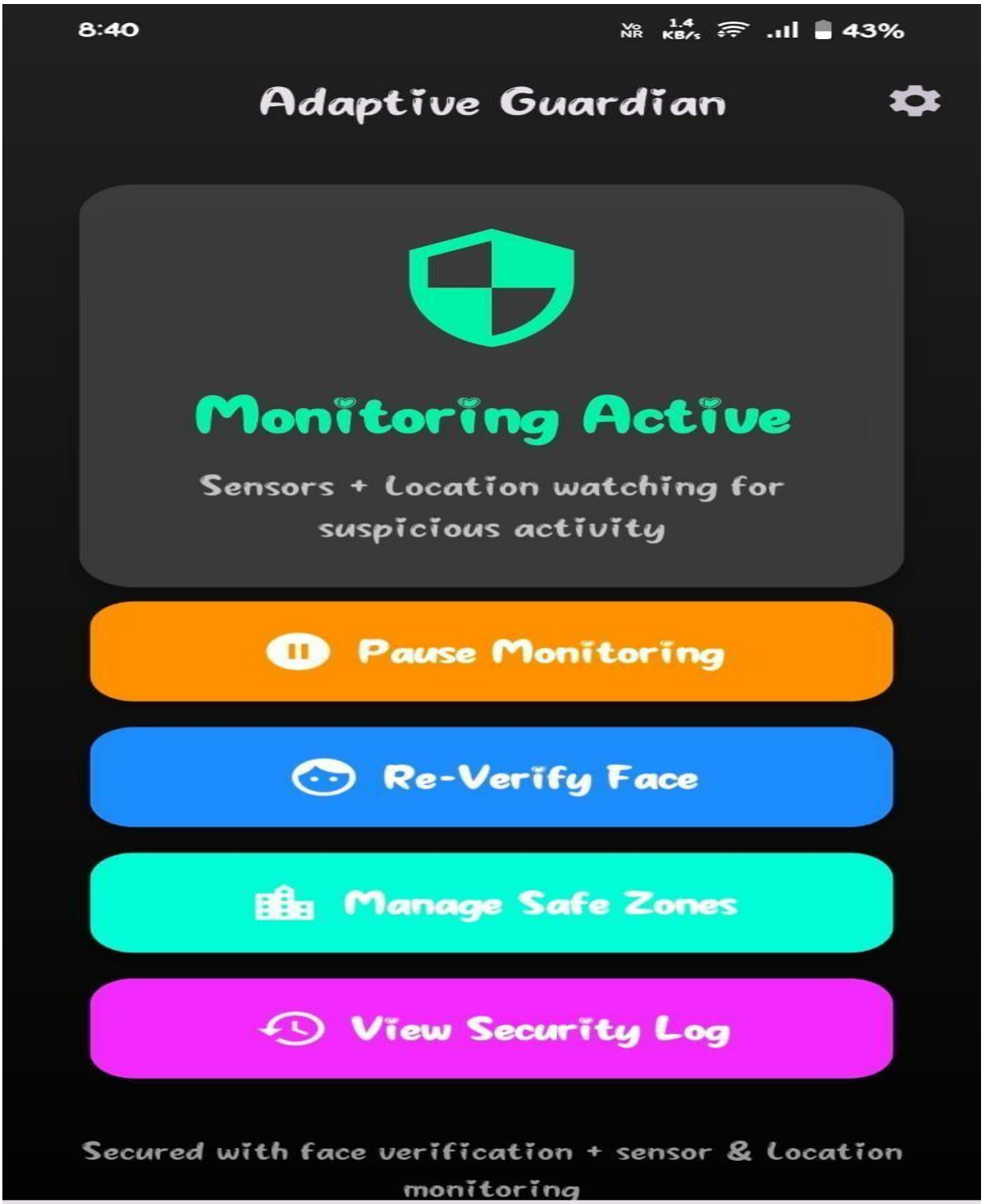

SECURITY-MOBILE-SELFLEARNI...
.dart_tool
version
.vscode
settings.json
android
.gradle
8.13
buildOutputCleanup
buildOutputCleanup.lock
cache.properties
outputFiles.bin
noVersion
vcs-1
file-system.probe
.kotlin\sessions
app
build\reports\problems
problems-report.html
gradle\wrapper
gradle-wrapper.jar
gradle-wrapper.properties
.gitignore
build.gradle.kts
gradle.properties
gradlew

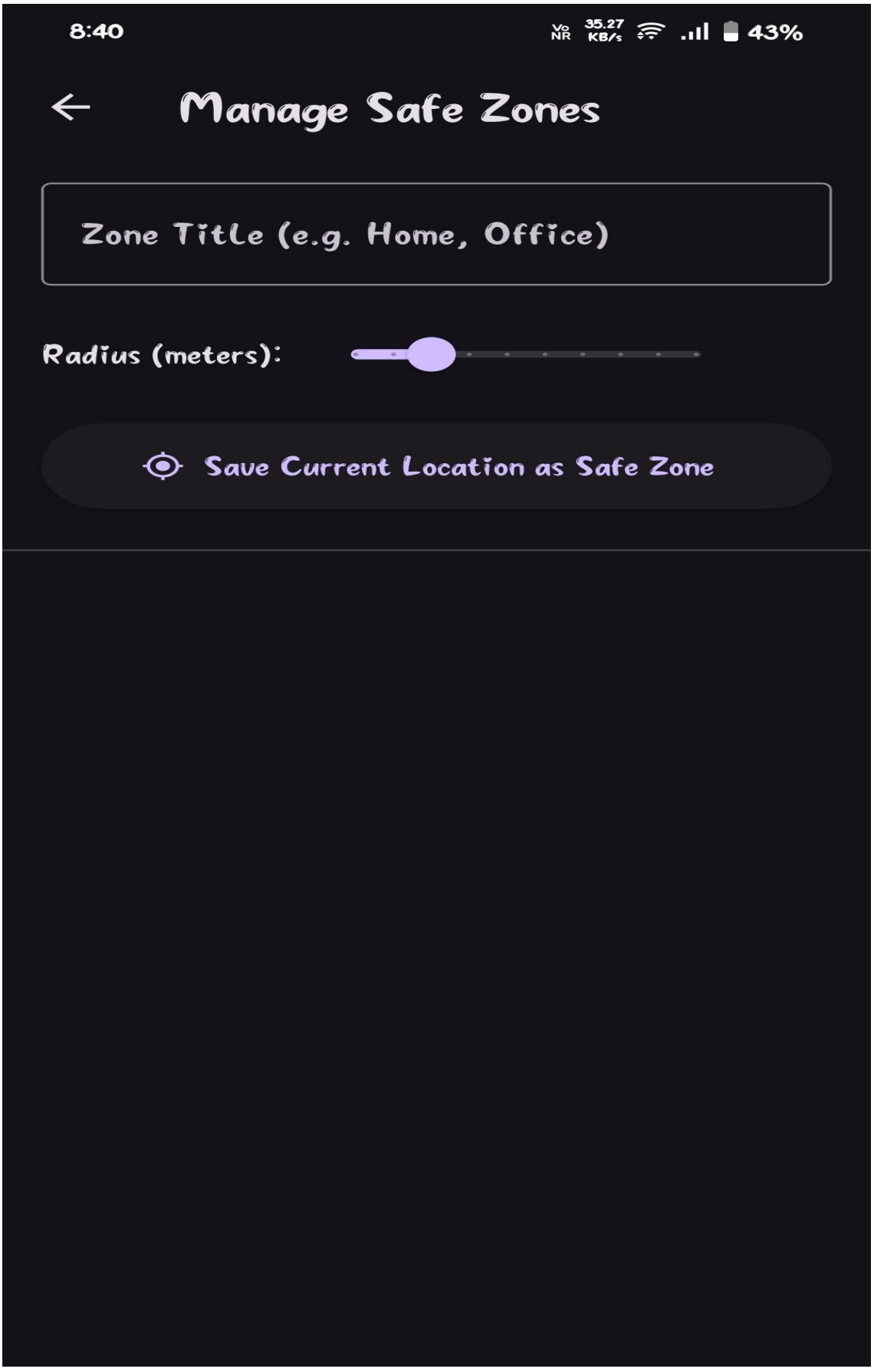

SECURITY-MOBILE-SELFLEARNI...
android
.gitignore
build.gradle.kts
gradle.properties
gradlew
gradlew.bat
local.properties
settings.gradle.kts
assets
mobilefacenet.tflite
build
ios
lib
linux
macos
test
web
windows
.flutter-plugins-dependencies
.gitignore
.metadata
analysis_options.yaml
pubspec.lock
pubspec.yaml
README.md

7. OUTPUT:


self_Learnin...


8:39
43%
Adaptive Guardian
Protected
Monitoring is currently paused
Start Monitoring
Re-Verify Face
Manage Safe Zones
View Security Log
Secured with face verification + sensor & location monitoring
Keep the app running for full protection


8:40
43%
Adaptive Guardian
Monitoring Active
Sensors + Location watching for suspicious activity
Pause Monitoring
Re-Verify Face
Manage Safe Zones
View Security Log
Secured with face verification + sensor & Location monitoring
Monitoring started


8:40
43%
Manage Safe Zones
Zone Title (e.g. Home, Office)
Radius (meters):
Save Current Location as Safe Zone

8. PROJECT CONCLUSION:

The project “Self-Learning Mobile Security System Using Face Detection and Behavioural Analysis with Artificial Intelligence (AI)” successfully presents an advanced and intelligent approach to mobile security. In an era where smartphones store vast amounts of sensitive information, relying solely on traditional authentication methods is no longer sufficient. This project addresses these challenges by integrating biometric authentication with behavioral analysis to provide a more secure and adaptive system.

By implementing face detection and recognition, the system ensures quick and contactless user verification. The addition of behavioral analysis further strengthens security by continuously monitoring user interaction patterns such as typing speed, touch behavior, and app usage. This dual-layer authentication mechanism significantly reduces the chances of unauthorized access.

One of the key achievements of this system is its self-learning capability. Using artificial intelligence and machine learning techniques, the system adapts to the user’s behavior over time, improving accuracy and minimizing false alarms. This dynamic nature makes the system more efficient compared to static security solutions.

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